

RETAIL SALES INTELLIGENCE

Where the profit is going

An analysis of 9,993 transactions · Jan 2015 – Dec 2018

\$2.30M revenue **\$286,409 profit** **12.47% margin**

The bottom line

One finding accounts for more profit than any growth initiative under consideration.

Discounts above 30% destroy \$125,007 in profit

That is 44% of everything the business earns — from just 11.7% of order lines.

 **\$2.30M**

REVENUE

+20.4% vs 2017

 **\$286,409**

PROFIT

12.47% margin

 **5,009**

ORDERS

793 customers

 **\$125,007**

PROFIT LOST

to deep discounts

Profit margin by discount band — the threshold sits at 30%.

A bar chart illustrating the distribution of discount preferences among respondents. The y-axis represents the percentage, ranging from -140% to 60% in increments of 20%. The x-axis lists five discount categories. The bars for 'No discount', '1-15%', and '16-30%' are green, while the bars for '31-50%' and '50%+' are red. The values are explicitly labeled above each bar.

Discount Category	Percentage
No discount	29.5%
1-15%	12.8%
16-30%	9.2%
31-50%	-24.8%
50%+	-119.2%

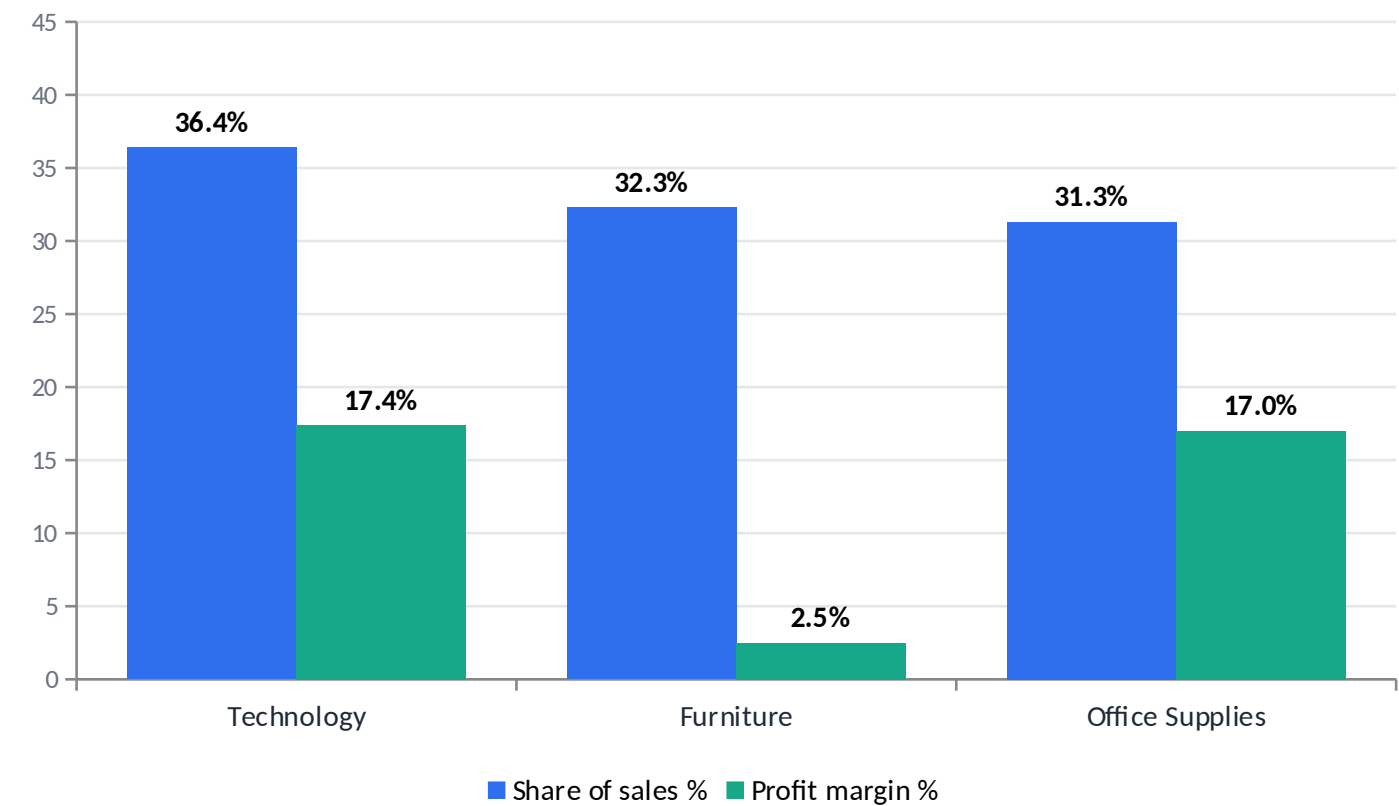
-119%

Each sale costs more than it earns

Query 14, *analysis_queries.sql*. Bands cut at 0 / 15 / 30 / 50 per cent.

Furniture is a third of revenue and none of the profit

Share of sales against realised margin, by category.



Two sub-categories lose money outright

 **-\$17,725**

TABLES

26% average discount — deepest in catalogue

 **-\$3,473**

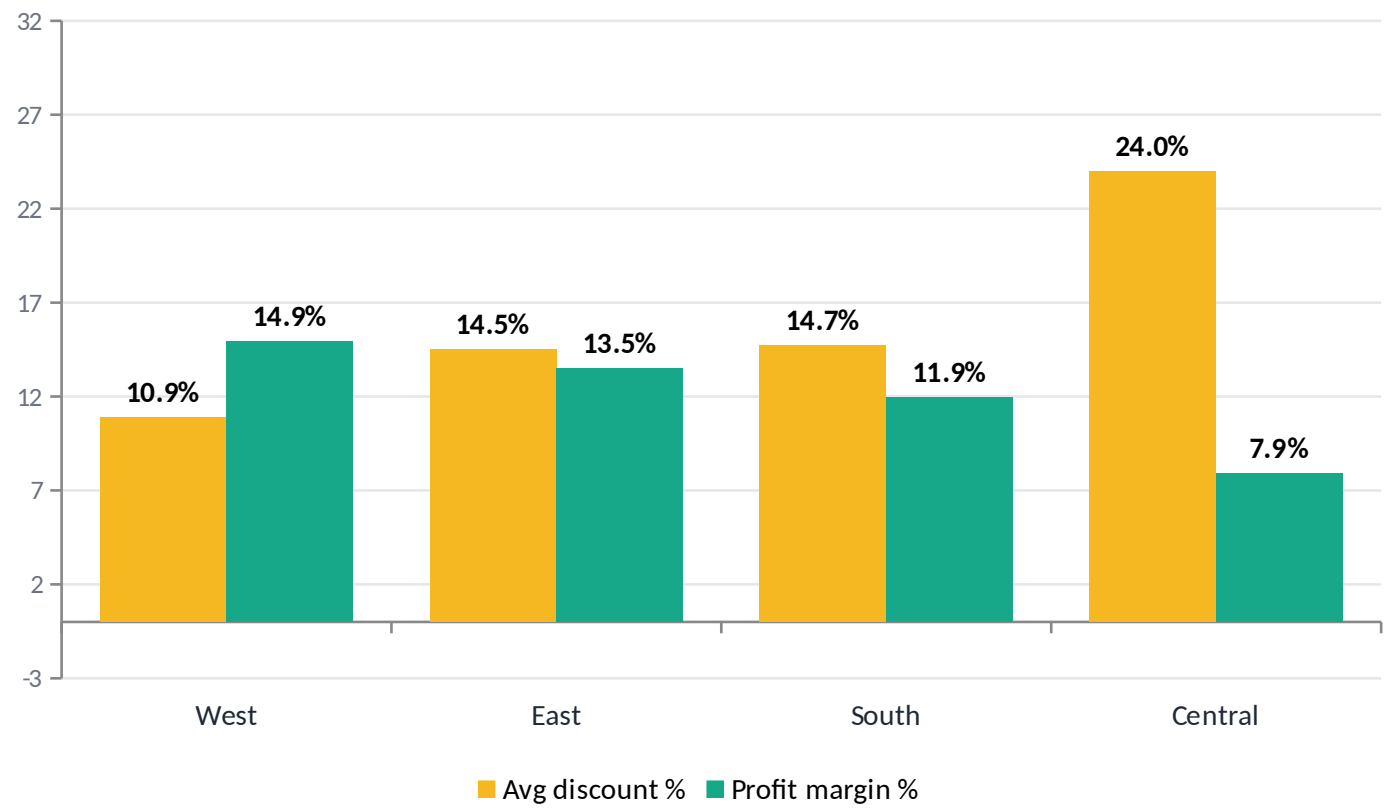
BOOKCASES

21% average discount

Same root cause as slide 3.

The geography of the problem is the same problem

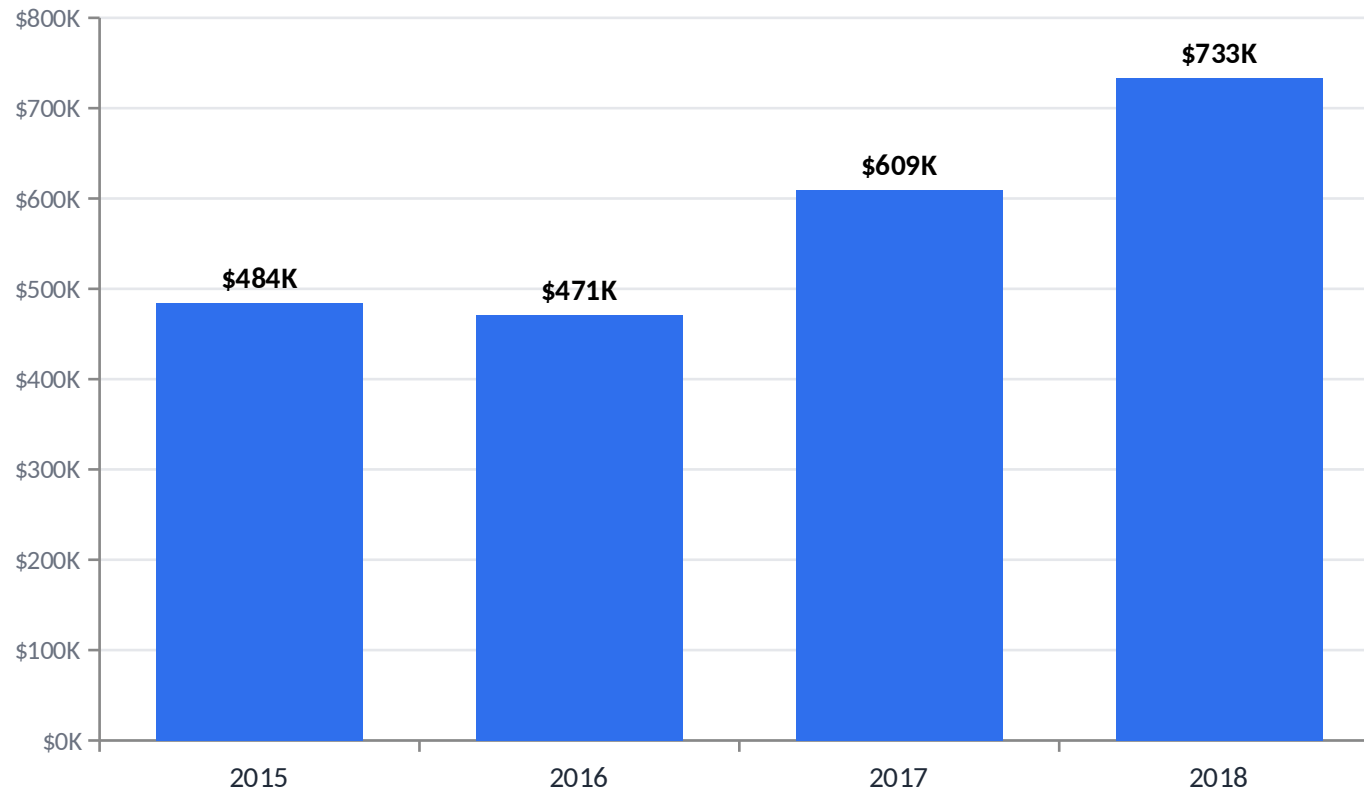
Every loss-making region and state discounts far harder than the profitable ones.



- 1 Central discounts twice as hard as West**
24.0% vs 10.9% — and earns half the margin.
- 2 10 states operate at a loss**
All discount between 28% and 40%.
- 3 Philadelphia is 5th by revenue and loses \$13,838**
Revenue rank and profit contribution are not the same thing.

The business is growing — that is not the issue

Annual revenue, with margin improving every year.



 **+20.4%**

2018 GROWTH

after +29.5% in 2017

 **10.2% → 12.7%**

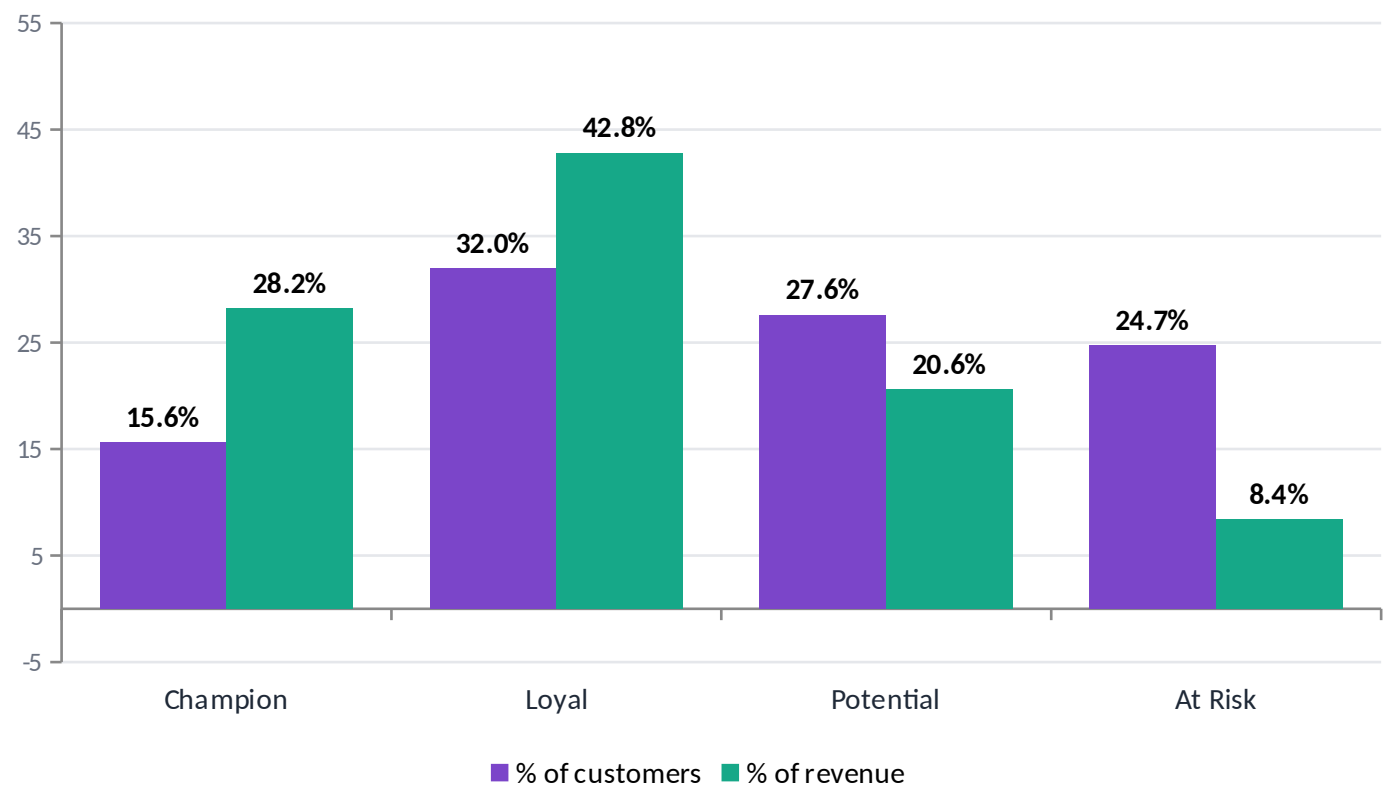
MARGIN, 2015 TO 2018

improving, but well below potential

Growth is not being bought with discounts in aggregate — the damage is concentrated in a narrow band of deep discounts.

Where the customer value sits

RFM segmentation of all 793 customers by recency, frequency and monetary value.



1 Champions and Loyal are 71% of revenue

378 accounts. This is the retention list.

2 At Risk is 197 customers, 8.4% of revenue

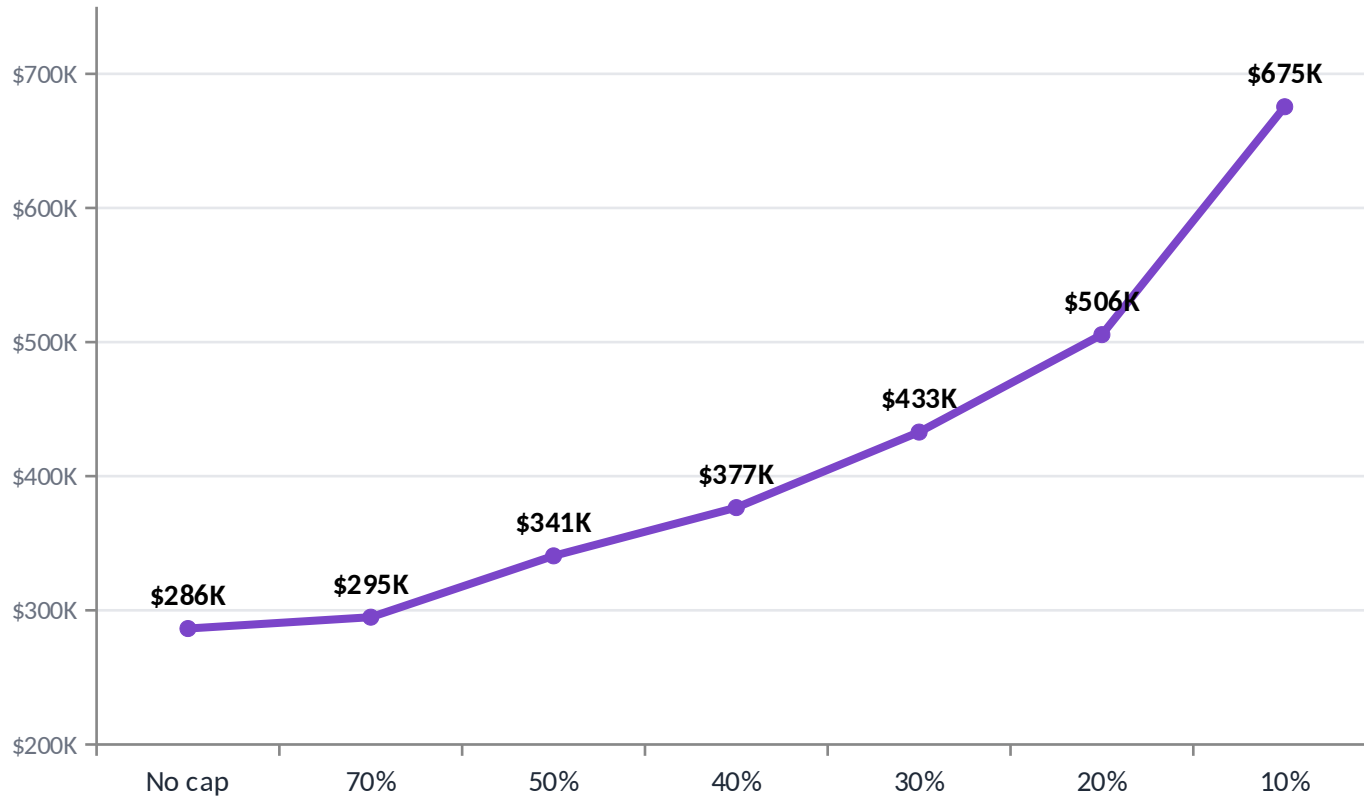
Average 329 days since last order — automate, don't staff it.

3 Top 20% produce 48% of revenue

Real concentration, but not the 80/20 usually assumed.

What a discount cap is worth

Simulated total profit at each maximum discount level.



\$432,855

PROFIT AT A 30% CAP

+51% against today

Why 30% and not lower

- The curve flattens at 30%. Tightening further returns less per order affected while touching far more of the book.
- A 30% rule affects one line in nine. A 20% rule affects nearly half.

Upper bound: assumes each sale still closes at the lower discount. Some would not — see method.

Recommendations

Ordered by value against effort. The first one carries most of the benefit.

- 1

Require approval for any discount above 30%
Worth +44% to +51% of profit. Affects 11.7% of order lines. Pilot in Central first.
- 2

Reprice or discontinue Tables and Bookcases
Removes a \$21K annual loss. Tables carry the deepest discounts in the catalogue.
- 3

Audit pricing authority in Central region
24.0% average discount against West's 10.9%, for half the margin.
- 4

Review the 10 loss-making states and Philadelphia
Philadelphia is 5th by revenue and loses \$13,838. Revenue rank is hiding it.
- 5

Weight inventory and campaign spend to Sep-Dec
November and December peak in all four years; January and February trail.

 **\$125,007**

AT STAKE
recoverable profit

 **1 in 9**

ORDER LINES AFFECTED
by the 30% rule

Method and limitations

What was done, and what these numbers cannot tell you.

How it was built

- 10,800 raw rows cleaned to 9,993: removed 806 empty records and 504 duplicates.
- 26 engineered features; star schema in SQL with 30 analysis queries.
- Totals reconciled independently across Python, SQL and Excel.
- 15 of 15 automated validation rules pass.

What it cannot tell you

- Correlation, not causation. Deep discounts coincide with losses; the data cannot prove the discount caused it rather than both following from clearing slow stock.
- The cap simulation assumes every sale still closes at the lower discount. Some customers would walk, so treat it as an upper bound.
- No COGS column, so margin uses the supplied profit figure at face value.
- The 3-month forecast has 18.8% backtest error — usable for direction, not for committing budget.

Recommended next step: run the 30% cap as a 90-day pilot in Central region and measure realised margin against the control regions. That converts an estimate into a decision.

IN ONE LINE

The business does not have a growth problem.

It has a discounting problem — and it is worth \$125,007 a year to fix.

Full analysis, SQL and reproducible pipeline: github.com/YOUR-USERNAME/retail-sales-intelligence-dashboard